

Shad Humydee

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Value Proposition

Machine Learning focused Data Science Leader **7+ yrs** specializing in Agentic AI, Recommender Systems, and NLP. Led and scaled production ML systems to **10M+ users**. Driving architecture, technical strategy, and cross-functional execution for full-stack ML platforms, translating AI into business-critical outcomes.

Professional Experience

Senior Data Scientist

Optimizely

Dec 2023 – Present

Hybrid, UK Team

- Architected and deployed **OPAL Evals**, a production-grade AI evaluation framework scoring conversational agent outputs using multi-model **LLM-as-a-Judge (Gemini, Claude)** across **pre-set** and **user-defined** custom quality dimensions. Designed a stateless evaluation engine API, CI/CD pipeline integration (**GitHub Actions**) with threshold-based release gating system for OPAL (**Optimizely's agent orchestration platform**), and an internal experimentation UI — reducing agent debugging time from **2–3 days to under 4 hours** and enabling **5x faster** prompt iteration cycles.
- Architected **Synapse**, a hybrid AI agent memory system for **OPAL**, enabling personalized AI experiences through long-term memory and contextual intelligence. Built async microservices using **FastAPI + Python** with real-time webhook ingestion, processing **10K+ agent interactions/day**.
- Engineered **CMAB recommendation engine** using contextual multi-armed bandits and XGBoost, handling **5K requests/second** for real-time ML predictions. Secured a major US financial services client with tens of millions of customers, reducing regret by **15%** and boosting conversions by **8%**.
- Built **AI-Backend**, a production-grade multi-tenant API gateway routing requests across **Azure OpenAI, Google Vertex AI/Gemini, and Replicate**. 3-layer architecture with streaming support, token-bucket rate limiting, and GKE deployment (**10 replica pods**, auto-scaling) with **Datadog APM** observability.

Software Engineer II (AI/ML)

Infolytx Inc.

Jan 2023 – Nov 2023

Hybrid, Dhaka

- Led architecture and delivery of the **Avicenna Healthcare NLP platform** using **spaCy, SciSpacy, AWS Comprehend Medical**, enabling physicians and insurance providers to process **500K+ patient records**, cutting assessment time by **50%**.
- Conducted PoCs and R&D integrating NLP/ML techniques with **AWS (DynamoDB, S3, EC2)** for healthcare applications, improving decision-making speed and accuracy across hospital networks.
- Mentored **2–3 junior engineers**, scaled team capability by **30%**, and established ML delivery best practices and code review standards.

Software Engineer I (AI/ML)

Infolytx Inc.

Jan 2022 – Dec 2022

Hybrid, Dhaka

- Shipped healthcare ML product from demo to MVP using **scikit-learn, TensorFlow**, reducing development-to-deployment cycle by **60%** and enabling real-time patient risk scoring.
- Built scalable ML/NLP pipelines with **Python, MySQL, AWS SDK (boto3)** processing **100K+ EMR records/day**, improving clinician workflow efficiency.
- Ensured **HIPAA compliance** through security best practices, peer code reviews, and production-ready ML workflow documentation.

Junior Software Engineer (AI/ML)

Infolytx Inc.

Jul 2021 – Dec 2021

Hybrid, Dhaka

- Built NLP solutions on EMR data using **NLTK, spaCy, NegSpacy**, improving clinical insight extraction accuracy by **25%** and reducing physician manual review workload.
- Enhanced ML model reliability via preprocessing, error handling, and **AWS cloud integration**, reducing pipeline failures by **30%**.

Machine Learning Engineer

Apurba Technologies Ltd.

Mar 2021 – Jun 2021

Remote, Dhaka Team

- Developed OCR preprocessing pipelines for Bangla language using **OpenCV, Pillow, and Cycle-GAN**, improving image-to-text conversion accuracy by **29%** as part of a national digitization initiative.
- Built predictive ML models using **scikit-learn, pandas, NumPy** processing **10K+ daily data points** for client analytics and decision-making.

Earlier Experience

Senior Business Analyst & Business Analyst <i>Integrated Software and Technologies Ltd.</i>	<i>Aug 2018 – Feb 2021</i> <i>Dhaka, Bangladesh</i>
- Secured high-ticket contracts including a \$2.5M project win, preparing technical proposals for clients including IOM, UNDP, UNODC, WFP and government bodies. - Led technical documentation and stakeholder coordination for complex government and enterprise systems, reducing onboarding time by 35% and project delays by 20%.	

Technical Skills

Agentic Workflows & Orchestration LangChain, RAG Architectures, LLM-as-Judge, Prompt Engineering, Tool Calling, Multi-step Reasoning	Python, TypeScript, Java
LLM APIs OpenAI (GPT-4o, Embeddings, DALL-E), Google Gemini, Anthropic (via Vertex AI)	AI-Assisted Engineering Claude Code, GitHub Copilot, Cursor
Machine Learning XGBoost, TensorFlow/Keras, Scikit-Learn, Transformers (BERT, T5), CMAB, Recommender Systems, SHAP/LIME	Architecture Patterns Event-Driven, Microservices, API Gateway
Data Science NumPy, Pandas, spaCy, SciSpacy, NegSpacy, NLTK, TensorBoard, Matplotlib, Seaborn	Cloud & Infrastructure (GCP-Focused) Vertex AI, GKE, Compute Engine, Pub/Sub, Cloud Spanner, Cloud Run, Scheduler, Storage, BigQuery, Datastore, Docker, Terraform
ML Ecosystem MLflow, Weights & Biases	Databases & Storage MySQL, PostgreSQL, Redis, DynamoDB, BigQuery
Languages & Frameworks	CI/CD GitHub Actions, Bitbucket Pipelines, Jenkins, Groovy
	Logging & Monitoring Datadog

Education

MSc in Computer Science and Engineering <i>United International University, Bangladesh</i> Focus: Machine Learning, Deep Learning, Natural Language Processing	<i>Class of 2026</i>
BSc in Naval Architecture and Marine Engineering <i>Military Institute of Science and Technology, Bangladesh</i> computational methods	<i>Jan 2014 - Dec 2018</i> Interdisciplinary foundation in engineering, mathematics, and

Key Honors and Awards

Judge, Bangla LLM Mathematical Reasoning Challenge, BUET CSE – Evaluated 15+ teams building Bangla language models for mathematical reasoning, assessing model architecture, prompt engineering, and benchmark performance. World Finalist, European Rover Challenge 2016, Poland — Ranked 22nd globally	<i>Nov 2024</i> <i>Sep 2016</i>
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Certifications

Fundamentals of Building AI Agents, IBM Claude Code in Action, Anthropic Introduction to Agent Skills, Anthropic Diagnosing Application Bugs with Datadog APM, Datadog Veracode Security Labs Champion Level 1, Veracode Introduction to PCI DSS for Developers, Veracode Software Security Awareness - OWASP Top 10 2021, Veracode	<i>Mar 2026</i> <i>Apr 2026</i> <i>Apr 2026</i> <i>Oct 2025</i> <i>Sep 2025</i> <i>Mar 2025</i> <i>Mar 2025</i>
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Languages

- English** — Native/Bi-Lingual Proficiency (daily collaboration with UK/US teams)
- Bengali** — Native/Bi-Lingual Proficiency